PROBLEM SET 5 DUE 4.30

1. Let $F : \mathscr{A} \to \mathscr{B}$ be a functor where $\mathscr{A}, \mathscr{B}$ are small categories. Prove the following:

(1) F has a right adjoint if and only if for any $Y \in \mathscr{B}$, the functor

$$\mathrm{Hom}_{\mathscr{B}}(F(-), Y) : \mathscr{A}^{\mathrm{op}} \to \mathbf{Set}$$

is representable.

(2) If the right adjoint of F exists, then it is unique up to natural isomorphism.

2. Find the left adjoint of the forgetful functor $U : \mathbf{CRing} \to \mathbf{Set}$.

3. Let $(F, G, \alpha) : \mathscr{A} \to \mathscr{B}$ be an adjunction. For any $X \in \mathscr{A}$ and $Y \in \mathscr{B}$, define ϵ_Y and η_X as follows:

$$\begin{aligned} \alpha_{G(Y),Y} : \mathrm{Hom}_{\mathscr{B}}(FG(Y), Y) &\longrightarrow \mathrm{Hom}_{\mathscr{A}}(G(Y), G(Y)) \\ \epsilon_Y &\longmapsto \mathrm{id}_{G(Y)} \\ \alpha_{X,F(X)} : \mathrm{Hom}_{\mathscr{B}}(F(X), F(X)) &\longrightarrow \mathrm{Hom}_{\mathscr{A}}(X, GF(X)) \\ \mathrm{id}_{F(X)} &\longmapsto \eta_X \end{aligned}$$

Show that

(1) $\epsilon = (\epsilon_Y)_Y$ is a natural transformation from FG to $\mathrm{id}_{\mathscr{B}}$

(2) $\eta = (\eta_X)_X$ is a natural transformation from $\mathrm{id}_{\mathscr{A}}$ to GF.

(3) $\epsilon F \circ F\eta = \mathrm{id}_F$ and $G\epsilon \circ \eta G = \mathrm{id}_G$.

The ϵ and η are called the **counit** and **unit** of (F, G, α) respectively. Find the units and counits of the following adjunctions:

(1) $- \otimes_A N$ and $\mathrm{Hom}_A(N, -)$ where N is an A-module;

(2) $B \otimes_A -$ and $-|_A$ where $f : A \to B$ is a ring map;

(3) $-|_A$ and $\mathrm{Hom}_A(B, -)$ where $f : A \to B$ is a ring map;

(4) the free functor $F : \mathbf{Set} \to A\textbf{-Mod}$ and the forgetful functor $U : A\textbf{-Mod} \to \mathbf{Set}$.

4. Let $F : \mathscr{A} \to \mathscr{B}$ and $G : \mathscr{B} \to \mathscr{A}$ be functors. Let $\epsilon : FG \Rightarrow \mathrm{id}_{\mathscr{B}}$ and $\eta : \mathrm{id}_{\mathscr{A}} \Rightarrow GF$ be natural transformations satisfying $\epsilon F \circ F\eta = \mathrm{id}_F$ and $G\epsilon \circ \eta G = \mathrm{id}_G$. Show that (F, G) is an adjoint pair via

$$\begin{aligned} \alpha_{X,Y} : \mathrm{Hom}_{\mathscr{B}}(F(X), Y) &\longrightarrow \mathrm{Hom}_{\mathscr{A}}(X, G(Y)) \\ \varphi &\longmapsto G(\varphi) \circ \eta_X \\ \epsilon_Y \circ F(\psi) &\longleftarrow \psi \end{aligned}$$

5. Choose *one* of the following statements to prove:

(1) If the sequence of A-modules

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \tag{a}$$

satisfies the property that for any A-module P, the sequence

$$0 \longrightarrow \mathrm{Hom}_A(P, M') \longrightarrow \mathrm{Hom}_A(P, M) \longrightarrow \mathrm{Hom}_A(P, M'')$$

is exact, then (a) is exact.

(2) If the sequence of A-modules

$$M' \longrightarrow M \longrightarrow M'' \longrightarrow 0 \tag{b}$$

satisfies the property that for any A-module Q, the sequence

$$0 \longrightarrow \mathrm{Hom}_A(M'', Q) \longrightarrow \mathrm{Hom}_A(M, Q) \longrightarrow \mathrm{Hom}_A(M', Q)$$

is exact, then (b) is exact.

6. Let $F : A\textbf{-Mod} \to B\textbf{-Mod}$ be an additive functor. Let M_1, M_2 be A-modules, let

$$i_1 : M_1 \longrightarrow M_1 \oplus M_2 \quad \text{and} \quad i_2 : M_2 \longrightarrow M_1 \oplus M_2$$

be the canonical injections, and let

$$p_1 : M_1 \oplus M_2 \longrightarrow M_1 \quad \text{and} \quad p_2 : M_1 \oplus M_2 \longrightarrow M_2$$

be the canonical projections. Let

$$\xi : F(M_1) \oplus F(M_2) \longrightarrow F(M_1 \oplus M_2)$$

be the B-module homomorphism induced by

$$F(i_1) : F(M_1) \longrightarrow F(M_1 \oplus M_2) \quad \text{and} \quad F(i_2) : F(M_2) \longrightarrow F(M_1 \oplus M_2)$$

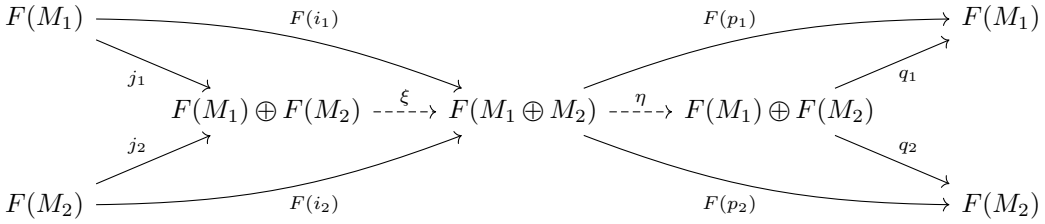
using the universal property of $F(M_1) \oplus F(M_2)$ as a direct sum. Let

$$\eta : F(M_1 \oplus M_2) \longrightarrow F(M_1) \oplus F(M_2)$$

be the B-module homomorphism induced by

$$F(p_1) : F(M_1 \oplus M_2) \longrightarrow F(M_1) \quad \text{and} \quad F(p_2) : F(M_1 \oplus M_2) \longrightarrow F(M_2)$$

using the universal property of $F(M_1) \oplus F(M_2)$ as a direct product.



Show that $\xi\eta = \mathrm{id}$ and $\eta\xi = \mathrm{id}$. In particular, ξ, η are B-module isomorphisms, and so

$$F(M_1 \oplus M_2) \cong F(M_1) \oplus F(M_2)$$

as B-modules.

7. Let $F : A\textbf{-Mod} \to B\textbf{-Mod}$ be an additive functor. Choose *one* of the following statements to prove:

(1) F is left exact if and only if for every short exact sequence of A-modules

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0,$$

the sequence of B-modules

$$0 \longrightarrow F(M') \longrightarrow F(M) \longrightarrow F(M'')$$

is exact.

(2) F is right exact if and only if for every short exact sequence of A-modules

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0,$$

the sequence of B-modules

$$F(M') \longrightarrow F(M) \longrightarrow F(M'') \longrightarrow 0$$

is exact.

8. Let $F : A\textbf{-Mod} \to B\textbf{-Mod}$ be an additive functor. Show that the following are equivalent:

(1) F is exact.
(2) F is left exact and preserves surjective homomorphisms.
(3) F is right exact and preserves injective homomorphisms.
(4) F transforms short exact sequences to short exact sequences.
That is, for every short exact sequence of A-modules

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0,$$

the sequence of B-modules

$$0 \longrightarrow F(M') \longrightarrow F(M) \longrightarrow F(M'') \longrightarrow 0$$

is exact.
(5) F transforms exact sequences to exact sequences.
That is, for every exact sequence of A-modules

$$M' \longrightarrow M \longrightarrow M'',$$

the sequence of B-modules

$$F(M') \longrightarrow F(M) \longrightarrow F(M'')$$

is exact.

9. Let $F : A\textbf{-Mod} \to B\textbf{-Mod}$ be an *exact* functor. Show that the following are equivalent:

(1) F is faithful.
(2) F reflects zero modules. That is, if $M \in A\textbf{-Mod}$ and $F(M) = 0$, then $M = 0$.
(3) F reflects zero morphisms. That is, if $f : M \to M'$ is A-linear and $F(f) = 0$, then $f = 0$.
(4) F reflects exact sequences. That is, if $M' \to M \to M''$ is a sequence in $A\textbf{-Mod}$ such that $F(M') \to F(M) \to F(M'')$ is exact in $B\textbf{-Mod}$, then $M' \to M \to M''$ is exact in $A\textbf{-Mod}$.

We say that an additive functor $F : A\textbf{-Mod} \to B\textbf{-Mod}$ is **faithfully exact** if it is exact and satisfies any of the above properties.

10. Let $(F, G, \alpha) : A\textbf{-Mod} \longrightarrow B\textbf{-Mod}$ be an adjunction of additive functors.
Prove the following statements:

(1) For any A-module M and B-module N, the set map

$$\alpha_{M,N} : \mathrm{Hom}_B(F(M), N) \longrightarrow \mathrm{Hom}_A(M, G(N))$$

is an isomorphism of abelian groups.
(2) F is right exact and G is left exact.
Hint: use Problem **7***.*

11. We discussed in lecture the definition of a pullback diagram and mentioned a proposition about some of their properties in a module category. We also mentioned that there is a dual notion of pushout diagram given by reversing the arrows. In this exercise, you will spell out this dual version in detail.

(1) State the definition of a pushout diagram in a category $\mathscr{C}$ and formulate a proposition about pushout diagrams analogous to the one stated on p.3-4 of the slides.
(2) Prove the proposition you formulated in (1). Alternatively, you may prove its dual version (about pullbacks) given on p.3-4 of the slides.